MDF SIGNAL CLEANING TOOLS(*Generated using Copilot)*

Torque Cleaner and Combined AUTO/CUSTOM Torque and Temperature Cleaner

DOCUMENT PURPOSE

A practical user guide and technical reference for running, reviewing, and validating the two Python-based MDF cleaning tools. The document reflects the supplied implementation and its actual folder, mode, channel, cleaning, logging, and output behavior.

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Application area	Vehicle measurement signal cleaning
Input formats	MDF, MF4, DAT
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Document Control

Item	Description
Document title	MDF Signal Cleaning Tools: User and Technical Guide
Scope	Torque-only cleaner and combined AUTO/CUSTOM cleaner
Primary users	Engineers and analysts handling vehicle MDF/MF4/DAT measurements
Owner	Nerea Mulubrhan
Status	Working documentation based on supplied Python implementations
Important limitation	The exact production Python filenames and storage paths must be inserted where placeholders are shown.

Important use statement

Always validate the generated before-and-after plots and the edited MF4 before approving cleaned data for engineering use. The scripts apply automated rules, but an engineering review remains necessary.

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1. System Overview

Two related Python tools were developed for cleaning paired vehicle measurement signals while preserving all non-cleaned channels in the output measurement file.

Tool	Primary purpose	Operating modes	Typical signals
Tool 1	Dedicated torque cleaning	Automatic torque workflow	TQ_FL, TQ_FR, TQ_RL, TQ_RR
Tool 2	Combined torque and temperature cleaning	AUTO and CUSTOM	Standard torque channels or manually selected temperature pairs

1.1 Common Capabilities

- Recursively processes .mdf, .mf4, and .dat files in a selected root folder.
- Creates a separate output root folder with the suffix _Edited.
- Mirrors the original folder structure inside the output root.
- Preserves the original source folder and source files.
- Generates full-range and event-focused before-and-after PNG plots.
- Creates an Excel event log when cleaned faults are available.
- Writes edited measurements in MDF 4.10 format.
- Supports long Windows paths and UNC network shares when saving images.

1.2 High-Level Workflow

```
Run the Python .py file
        |
Select cleaning mode, when using Tool 2
        |
Select the root measurement folder
        |
Find MDF, MF4, and DAT files recursively
        |
Load selected signal pairs
        |
Detect, filter, and clean accepted faults
        |
Create plots, Excel log, and output MF4
        |
Review results in <Selected_Root>_Edited
```

2. Installation and Code Location

2.1 Recommended Project Layout

```
....\Personal_Folders\Mulubrhan_A\
Data_clean_substitute_interpolation_Torque_Only.py
```

2.2 Python Requirements

The scripts require Python 3 and the following packages or standard-library modules:

Category	Packages or modules
External packages	asammdf, numpy, matplotlib, pandas, openpyxl
Python standard library	tkinter, pathlib, shutil, os, re

```
pip install asammdf numpy matplotlib pandas openpyxl
```

2.3 Running in Spyder

1. Open Spyder.
2. Open the required .py file.
3. Select the Python environment where the required packages are installed.
4. Press F5 or select Run > Run.
5. Follow the console prompts and folder-selection dialog.

2.4 Running from a Command Prompt

```
cd C:\Path\To\....\Personal_Folders\Mulubrhan_A
python Data_clean_substitute_interpolation_Torque_Only.py
```

Folder selection

Both tools ask the user to select a root folder. They do not ask the user to select each measurement file individually.

3. Tool 1: Torque-Only Cleaner

3.1 Purpose and Input

The torque-only cleaner is a dedicated batch tool for detecting and correcting faults in paired wheel-torque signals. It searches the chosen root folder and all subfolders for supported measurement files.

Drive type	Axle processed	Channel pair
FWD	Front only	TQ_FL or Tq_FL, and TQ_FR or Tq_FR
RWD	Rear only	TQ_RL or Tq_RL, and TQ_RR or Tq_RR
AWD	Front and rear	All four standard torque channels

3.2 Drivetrain Detection

The script searches the complete file path for the text **FWD, RWD, or AWD**. Folder naming therefore controls which axle is processed.

```
Recommended structure:
Measurements\FWD\...
Measurements\RWD\...
Measurements\AWD\...
```

Important

If the path does not contain FWD, RWD, or AWD, both front and rear processing may be attempted because neither the FWD nor RWD skip condition is activated.

3.3 Processing Sequence

6. Load the MDF/MF4/DAT file with asammdf.
7. Identify the relevant torque channels.
8. Process rear and/or front axle according to drivetrain text in the path.
9. Detect candidate faults using paired-signal comparisons.
10. Reject events that resemble valid common vehicle behavior.
11. Clean accepted intervals.
12. Generate plots and the event log.
13. Save an edited MF4 if cleaning occurred, otherwise copy or convert the original file.

3.4 Torque Fault Types

Fault type	Detection concept	Correction
Point spike	Sharp isolated sample with strong deviation from local baseline and opposite signal	Same-signal interpolation or opposite-side substitution
Offset spike	Local disturbance with a small slope or multiple affected samples	Interpolate the expanded faulty interval
Short or thin fault	Brief pulse with clear deviation and return	Same-signal interpolation
Step or plateau	Sustained abnormal level with strong pair difference	Side-specific interpolation
Opposite-sign plateau	Both sides high with opposite signs	Clean faulty side or sides independently
Long stuck or frozen signal	Most samples remain close to a constant median level	Substitute from the valid opposite signal

3.5 Main AUTO Torque Thresholds

Setting	Value	Purpose
value_threshold	100 Nm	Starts paired-signal difference checking
window_size	20 samples	Detection window
activity_ratio	3.0	Requires one side to be much more dynamic
spike_jump	200 Nm	Sharp jump threshold
high_torque_threshold	1500 Nm	High-level plateau candidate
direct_step_abs_min	3000 Nm	Clear saturated step condition
step_mean_diff_threshold	500 Nm	Strong left-right difference
min_step_samples	5 samples	Minimum accepted step size
min_stuck_duration_s	20 s	Minimum duration for long stuck checking

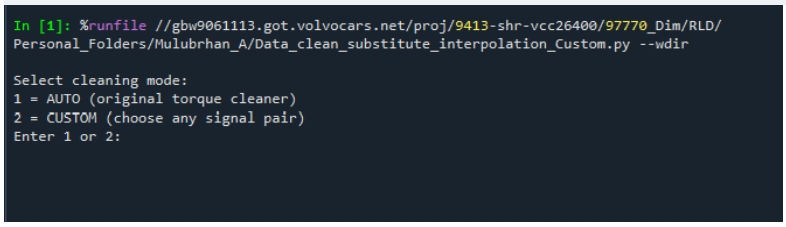
Engineering interpretation

These constants work together. A value exceeding one threshold is not automatically cleaned. Shape, slope, paired-signal difference, duration, and return-to-baseline checks are also used.

4. Tool 2: Combined AUTO/CUSTOM Cleaner

4.1 Mode Selection

```
Select cleaning mode:
1 = AUTO (original torque cleaner)
2 = CUSTOM (choose any signal pair)
Enter 1 or 2:
```



Entering 2 selects CUSTOM. Any other entry selects AUTO.

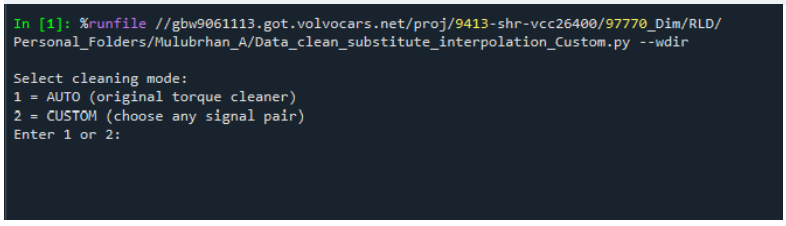
Mode	Behavior	Best use
AUTO	Uses the original fixed torque channels and path-based axle handling	Standard FWD, RWD, and AWD torque cleaning
CUSTOM	Uses manually selected paired channels and temperature-specific rules	Air or rubber temperature cleaning

4.2 AUTO Mode

AUTO mode preserves the original torque-cleaner behavior. Rear uses TQ_RL versus TQ_RR, and front uses TQ_FL versus TQ_FR. Drivetrain handling is based on FWD, RWD, or AWD text in each file path.

4.3 CUSTOM Drivetrain Selection

```
Select custom vehicle configuration:
1 = FWD
2 = RWD
3 = AWD
```



CUSTOM mode asks for the drivetrain manually. The chosen configuration is applied to every file in the selected batch. An invalid choice defaults to AWD.

Batch consistency rule

Use one CUSTOM batch only for files that share the selected configuration and channel names. The channels are selected from the first measurement file and reused for every file.

4.4 CUSTOM Channel Selection

Channel selection is performed from the first measurement file found in the root folder. A channel can be entered by its sorted-list index or by its exact name. Partial or prefixed matches can also be resolved when unique.

Configuration	Required selections	Example
FWD	Front left and front right	FL_Air, FR_Air
RWD	Rear left and rear right	RL_Rubber, RR_Rubber
AWD	Front left, front right, rear left, rear right	FL_Rubber, FR_Rubber, RL_Rubber, RR_Rubber

4.5 Active Temperature Rules

Rule	Current state	Key settings
Physical validity range	Enabled	15°C to 120°C
Short invalid values	Enabled	15-sample mask padding
Local temperature dips	Enabled	1.2°C drop, 10 s window, 3 s padding, 80th percentile baseline
Mostly invalid signal	Enabled	75% invalid; reference must be 20% invalid or less
Final remaining-dip repair	Disabled	Function exists but is not invoked

Current scope

Although the console says “choose any signal pair,” the active CUSTOM validity rules classify values outside 15 to 120 as invalid. CUSTOM mode should therefore be treated as temperature-oriented unless the rules are redesigned for another signal type.

5. Cleaning Logic and Algorithms

5.1 Short Invalid Temperature Values

Values below 15°C, above 120°C, NaN, or infinity are marked invalid. The mask is expanded, and stable valid medians before and after the event are used as interpolation boundaries. This catches dropouts such as -60°C while avoiding a ramp based on the first unstable value after the fault.

5.2 Local Temperature Dips

A local upper baseline is calculated using the 80th percentile in a 10-second window. A sample is a candidate when it is more than 1.2°C below that baseline. Detected regions are extended by 3 seconds and repaired using the same signal's actual valid boundary values.

Interpolation behavior

The local baseline is used for detection only. The cleaner does not force the signal up to the percentile baseline. It interpolates between real valid values from the same signal.

5.3 Mostly Invalid Temperature Signals

If at least 75% of one selected temperature signal is invalid and no more than 20% of the paired reference is invalid, the complete faulty signal is substituted from the paired signal.

```
Example:
RL_Rubber is mostly -60°C
RR_Rubber is mostly valid
Result: RL_Rubber_CLEAN receives the RR_Rubber values
```

5.4 Sample-Rate Adaptation

The script calculates the median timestamp interval and converts the 10-second window and 3-second padding into sample counts. This improves consistency across files with different sampling rates.

5.5 Torque Interpolation and Substitution

Situation	Preferred action	Fallback
Short spike with similar pre-event and post-event level	Interpolate within the same signal	Use opposite paired signal if interpolation is not reliable
Step or plateau	Interpolate the side-specific detected region	Clean the side with the larger abnormal level
Short plateau pulse	Interpolate across the pulse	None
Global stuck channel	Substitute from the valid paired signal	No change when both sides are too similar to identify a reference

5.6 False-Positive Protection

- Similar-shape and similar-slope checks reject events where both paired signals move together.
- A return-to-baseline check distinguishes isolated spikes from valid operating transitions.
- Normal high-load behavior is protected when both signals are high, have the same direction, and share a common shape.
- Left-side and right-side masks allow independent fault decisions.
- Long stuck intervals are protected from being processed again as step or plateau faults.
- Nearby detections are merged for plotting to reduce duplicate images.

5.7 Event Merging for Review

Event group	Merge rule for plotting
Step/plateau	Merge when the next event begins within 3 seconds of the previous event end
Short/thin	Merge when events are within 3 seconds
Spike	Create only one plot when cleaned spike events are within 3 seconds

6. Outputs, Logs, and Folder Structure

6.1 Edited Root Folder

```
Selected input root:
Vehicle_623

Generated output root:
Vehicle_623_Edited
```

The original relative subfolder structure is mirrored under the edited root.

6.2 Output Naming Rules

Processing result	Measurement output	Additional outputs
At least one fault cleaned	<FileStem>_Edited.mf4	<FileStem>_Edited_Log.xlsx and Images_Edited
No cleaned fault, source MF4	<FileStem>.mf4 copied unchanged	Full-range plots may be created
No cleaned fault, source MDF or DAT	<FileStem>.mf4 converted to MDF 4.10	Full-range plots may be created

6.3 Example Output

```
Vehicle_623_Edited\AWD\
  Measurement_01_Edited.mf4
  Measurement_01_Edited_Log.xlsx
  Images_Edited\
    Measurement_01~(t=_all)_1before.png
    Measurement_01~(t=_all)_2after.png
    Measurement_01_Rear~(t=125.4)_1before.png
    Measurement_01_Rear~(t=125.4)_2after.png
```

6.4 Edited MF4 Content

- All original channels not selected for replacement are retained.
- Cleaned signals are appended using their original channel names.
- The original signal unit is retained when available.
- A comment identifies the channel as a cleaned version.
- The final file is converted to MDF version 4.10.

6.5 Image Naming and Meaning

Suffix or pattern	Meaning
_1before.png	Original signal view before cleaning
_2after.png	Signal view after cleaning
~(t=_all)	Full measurement time range
ShortThin	Short or thin fault zoom
<Axle>~(t=<time>)	Axle-specific event zoom

6.6 Excel Event Log

For edited files, the Excel log is written before the final MF4 save. This means the log can remain available even if the MF4 save later fails.

Column	Meaning
file_name	Source measurement stem
axle	Front, Rear, or Processed
window_name	Generated after-image filename
error_type	StepPlateau, Spike, ShortThin, or All_Cleaned
start_time_s	Event start time in seconds
end_time_s	Event end time in seconds
image_path	Path to the generated image

6.7 Long Network Paths

Images are first written to a local temporary directory and then copied to the final UNC or long Windows path.

```
Temporary image directory:
C:\Temp\TorqueCleaner_Plots
```

7. Validation Procedure

Approval principle

A successful script run does not by itself prove that every correction is physically valid. Review the plots and the edited measurement before using the output.

7.1 File-Level Validation Checklist

14. Open the corresponding output subfolder under the _Edited root.
15. Confirm whether the measurement filename has the expected _Edited suffix.
16. Open the Excel log and identify the event type and time range.
17. Compare each 1before image with the matching 2after image.
18. Review the full-range before-and-after plots.
19. Load the edited MF4 in the normal measurement analysis environment.
20. Confirm that the cleaned interval is smooth and physically reasonable.
21. Confirm that valid events outside the fault interval were not changed.
22. Confirm that non-selected channels remain present.
23. For CUSTOM temperature substitution, verify that the selected reference channel is physically appropriate.

7.2 Review Questions

Question	Expected result
Was the fault actually present in the original signal?	Yes, and it is visible in the before plot or source measurement
Was only the faulty interval changed?	Yes, except full-signal substitution for a mostly invalid or stuck channel
Does the cleaned signal reconnect smoothly?	Yes, without an artificial jump or long unrealistic ramp
Was common physical behavior preserved?	Yes, paired movement should remain unchanged
Are units and channel names retained?	Yes, output uses original names and available units

8. Troubleshooting and Limitations

Console message or issue	Likely cause	Action
No folder selected. Script stopped.	Folder dialog was cancelled	Run again and select a root folder
No MDF/MF4/DAT files found	No supported files in the selected root	Check folder and file extensions
Front/Rear channels not available or failed	Selected channels are absent or unreadable	Verify channel names in every batch file
FAILED IMAGE SAVE	Path, permission, or local temp-folder issue	Check C:\Temp and output-folder write access
FAILED to save Excel log	Path, permission, or open workbook	Close the workbook and verify write access
FAILED to save cleaned full MF4	Output path, permissions, or MDF channel issue	Review console error and test on one local file
CUSTOM output looks incorrect	Signal is not a temperature signal or pair selection is wrong	Use temperature pairs and verify selected channels

8.1 Known Implementation Limitations

- CUSTOM mode always asks for FWD, RWD, or AWD manually.
- The CUSTOM drivetrain and selected channels apply to the complete batch.
- CUSTOM channel selection uses the first measurement file, so all files must contain compatible channels.
- CUSTOM mode includes generic step and spike logic in addition to temperature rules.
- The final remaining-temperature-dip function exists but is disabled and not called.
- AUTO drivetrain handling depends on path text.
- Thresholds are rule-based and require engineering validation on new vehicle or signal populations.

Safe batch practice

Test one representative measurement first, review all generated plots, and then scale the same configuration to the full folder.

9. Quick Operating Procedures

9.1 Torque-Only Cleaner

24. Open the torque-only Python file.
    .... \RLD\Personal_Folders\Mulubrhan_A
    Data_clean_substitute_interpolation_Torque_Only.py
25. Run the complete script.
26. Select the root folder containing vehicle measurement files.
27. Allow the script to detect FWD, RWD, or AWD from each path.
28. Wait for recursive processing to finish.
29. Open <RootFolder>_Edited.
30. Review edited MF4 files, Excel logs, and Images_Edited folders.

9.2 Combined Cleaner in AUTO Mode

31. Run the combined Python file.
32. Enter 1 for AUTO.
33. Select the root measurement folder.
34. Allow the standard torque channels to be processed.
35. Review the generated _Edited output root.

9.3 Combined Cleaner in CUSTOM Temperature Mode

36. Run the combined Python file.
37. Enter 2 for CUSTOM.
38. Select the root folder.
39. Choose FWD, RWD, or AWD.
40. Enter the required temperature channel names or channel indices.
41. Confirm the displayed selected channels.
42. Allow the batch to finish.
43. Review interpolation and any full-signal substitution carefully.

Recommended CUSTOM use

Use paired channels representing the same physical quantity and comparable locations, for example left and right air temperature or left and right rubber temperature.

Appendix A. Configuration Reference

A.1 CUSTOM Temperature Settings

Configuration variable	Current value	Meaning
CUSTOM_TEMPERATURE_RULE_ENABLED	True	Enable invalid temperature cleaning
CUSTOM_TEMP_MIN_VALID	15.0	Minimum valid temperature
CUSTOM_TEMP_MAX_VALID	120.0	Maximum valid temperature
CUSTOM_TEMP_PAD_SAMPLES	15	Padding around invalid values
CUSTOM_TEMP_LOCAL_DIP_RULE_ENABLED	True	Enable local dip cleaning
CUSTOM_TEMP_LOCAL_DROP_THRESHOLD	1.2	Minimum dip below local baseline
CUSTOM_TEMP_LOCAL_WINDOW_SEC	10.0	Time window for local baseline
CUSTOM_TEMP_LOCAL_DIP_PAD_SEC	3	Time padding around detected dips
CUSTOM_TEMP_LOCAL_BASELINE_PERCENTILE	80	Upper local baseline percentile
CUSTOM_TEMP_FINAL_DIP_RULE_ENABLED	False	Final repair is disabled
CUSTOM_TEMP_LONG_INVALID_RULE_ENABLED	True	Enable mostly invalid substitution
CUSTOM_TEMP_LONG_INVALID_RATIO	0.75	Faulty-signal invalid ratio
CUSTOM_TEMP_REFERENCE_MAX_INVALID_RATIO	0.20	Maximum invalid ratio in reference

A.2 Important Processing Constants

The code contains additional torque and adaptive CUSTOM thresholds. Changes should be tested on representative known-good and known-fault measurements before release.

Appendix B. Console Messages and Meaning

Message	Meaning
Selected MODE = AUTO/CUSTOM	Confirms the selected operating mode
Measurement files found: N	Number of supported files found recursively
sample rate = ... Hz	Estimated sample rate from timestamps
local temperature window = ... samples	10-second CUSTOM window converted to samples
local temperature padding = ... samples	3-second CUSTOM padding converted to samples
CUSTOM temp invalid ratio LEFT/RIGHT	Fraction outside the valid range
merged Step/Plateau plot events	Number of grouped step events for plotting
merged ShortThin plot events	Number of grouped short events
merged Spike plot events	Number of grouped cleaned spikes
Saved Excel log	Event workbook was written
Saved full edited MF4	Final edited MDF 4.10 file was written

Final Summary

The torque-only cleaner provides a dedicated rule-based workflow for standard wheel-torque channels. The combined cleaner retains that workflow in AUTO mode and adds a CUSTOM temperature-oriented mode with manual channel selection, time-based dip detection, short-invalid interpolation, and full-signal substitution when one paired temperature signal is mostly invalid. Both tools preserve the source folder, create an _Edited output root, and support visual and logged validation of every processing run.

Next maintenance step

When the Python scripts are renamed, moved, or their thresholds are updated, revise the document-control section, code-location section, and Appendix A so this guide remains aligned with the executable version.